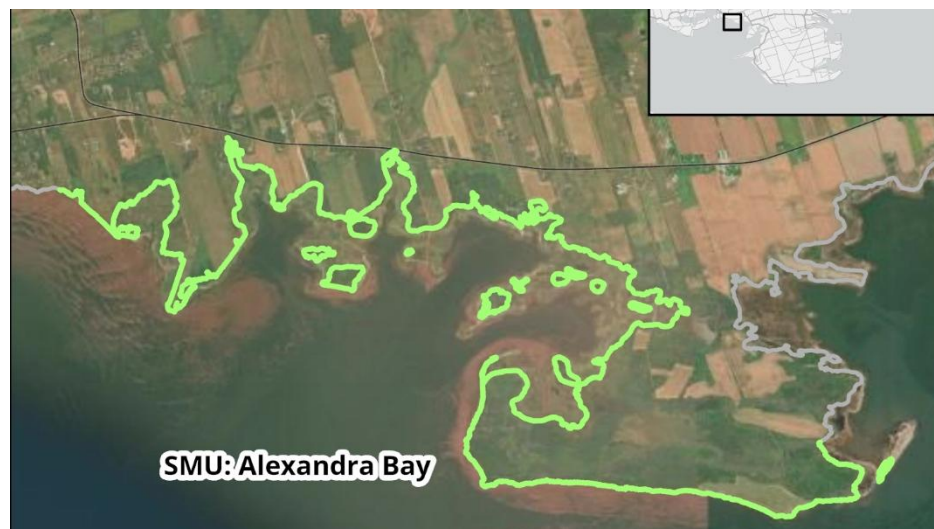


1. Alexandra Bay

Shoreline length: 30 km



Shore Types

Alexandra Bay is a bay located on the southeast end of PEI. Shore type characterized by a majority of wetlands, plus sloped armour along some properties in Stratford between Bellevue Point and Tea Hill Park.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
14	5	3	6	72	0	0	0

Governance, Infrastructure, and Economy

Approximately 73% of this area is within the Rural Municipality of Alexandra; 12% is unincorporated; the remaining 15% is in the Town of Stratford. There are no key coastal economic drivers within the coastal

section of this region. Local and legacy land use constraints include several small, developed lots on unpaved roads along the coast. No key infrastructure is noted.



Armoured shoreline on west end of SMU at Haszard Point, Stratford



Wetlands in Jardines Bay

Natural Assets and Heritage

Tea Hill Park has beach access and a cricket field and playground. The Alexandra Point Rifle Range is also located in this area. The area is considered critical habitat for the threatened Bank Swallow.

Planning and Policy Framework

Provincial rules apply in the Alexandra and unincorporated portions of this SMU (85% combined). Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood

elevations identified in regulations, but coastal hazard assessments are typically conducted. Stratford has municipal planning authority. Current planning documents apply watercourse and wetland regulations in all zones, including an O2 Reserve Zone — not shown on the zoning map — whose boundaries are interpreted to include any area defined as a Wetland or Watercourse, plus the area within 60 metres (197 feet) of a Wetland or Watercourse. Draft documents currently out for consultation would add vertical elevation requirements for habitable floors.

Sediment transport

The estimated net sediment transport rate averaged across the SMU is low.

Coastal Flood and Erosion Risk

Most of the low-lying shores are vulnerable to increasing flood hazards. Erosion hazard is highest along the exposed cliffs, bluffs, and low plains on the west end. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	2 [<1/km]	3 [<1/km]	6 [<1/km]	12 [<1/km]	1 [<1/km]	1 [<1/km]	1 [<1/km]
Roads	1766 m [60 m/km]	2097 m [71 m/km]	2447 m [83 m/km]	3279 m [111 m/km]	0	0	0

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
1.00 Alexandra Bay	NAI	MR	MR
Rationale	Given the initially low risk levels the proposed strategy is No Active Intervention. Mid- to long-term Managed Realignment will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including cliff and wetland habitat with increasing flood hazard.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.